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Inventor(s)	Corry; Michael et al.

Integrated power converter with isolated-channel feedback interface

Abstract

In some examples, an apparatus includes: a power stage having a power stage voltage input, a power stage control input, and a power stage output; a controller having a power stage control output coupled to the power stage control input, an output voltage status terminal, and a status signal input; a transformer having a primary side coil coupled to the power stage output, and a secondary side coil; a feedback voltage processor having a feedback input coupled to the secondary side coil, and a feedback output; a control message generator having a control input coupled to the secondary side coil, and a control message output; a status signal generator having first and second signal inputs coupled to respective feedback and control message outputs, and a status signal output; and a communication channel device coupled between the status signal input and output.

Inventors: Corry; Michael (Plano, TX), Formenti; Jose V. (Dallas, TX), Martinez; Robert (Lucas, TX)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

Family ID: 88849784

Assignee: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

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Primary Examiner: Rosario-Benitez; Gustavo A

Background/Summary

BACKGROUND

(1) An isolated power converter can provide galvanic isolation between circuits coupled to a voltage input of the power converter (e.g., a power source) and circuits coupled to a voltage output of the power converter (e.g., a holding capacitor and a load). The galvanic isolation can prevent current flowing directly between the input and output of the power converter, which can improve safety.

(2) An isolated power converter may include a power stage and a transformer. The primary side of the transformer can be coupled to the power stage and the power source, the secondary side of the transformer can be coupled to the load and the holding capacitor, which are coupled in parallel, and the transformer can provide the galvanic isolation. During an on-time period in which the power

stage is turned on/enabled, the load and the capacitor can be magnetically coupled to the power source through the transformer. The load can receive a current from the power source via the transformer, and the holding capacitor can be charged by the power source via the transformer to a target voltage. During an off-time period in which the power stage is turned off/disabled, the load and the holding capacitor can be disconnected from the power source, and the holding capacitor can discharge and supply a current to the load. The target voltage across the load/holding capacitor, and the current supplied to the load, can be determined based on the respective durations of the on-time period and the off-time period of the power stage.

SUMMARY

(3) An apparatus includes a power stage and a controller. The power stage has a power stage control input. The controller has a control input, a control output, and an output voltage status terminal. The control output is coupled to the power stage control input. The controller is configured to: receive a status signal at the control input, in which the status signal includes a feedback indicator and a control message, the feedback indicator represents a relationship between an output voltage and a target voltage, and the control message indicates whether the output voltage is within a target voltage range; responsive to the feedback indicator, provide a control signal at the control output; and responsive to the control message, set a state of the output voltage status terminal indicating whether the output voltage is within the target voltage range.

(4) An apparatus includes a feedback voltage processor, a control message generator, and a status signal generator. The feedback voltage processor has first and second inputs and a feedback output. The first input is coupled to a power converter output terminal. The second input is coupled to a target voltage terminal. The feedback voltage processor is configured to generate a feedback indicator at the feedback output responsive to a first voltage at the power converter output terminal and a target voltage at the target voltage terminal. Also, the control message generator has a control input and a control message output. The control input is coupled to the power converter output terminal. The control message generator is configured to provide a control message at the control message output responsive to whether the first voltage is within a target voltage range. Further, the status signal generator has a first signal input, a second signal input, and a status signal output. The first signal input is coupled to the feedback output. The second signal input is coupled to the control message output. The status signal generator is configured to: receive the feedback indicator at the first signal input; receive the control message at the second signal input; and provide a status signal at the status signal output, the status signal including the feedback indicator and the control message.

(5) An apparatus includes a primary side circuit, a secondary side circuit, a transformer, and a communication channel device. The primary side circuit includes a power stage and a controller. The power stage has a power stage voltage input, a power stage control input, and a power stage output. The controller has a power stage control output, an output voltage status terminal, and a status signal input, and the power stage control output is coupled to the power stage control input. The transformer has a primary side coil and a secondary side coil, and the primary side coil is coupled to the power stage output. The secondary side circuit has a secondary side input and an output voltage terminal, the secondary side input coupled to the secondary side coil. The secondary side circuit includes a feedback voltage processor, a control message generator, and a status signal generator. The feedback voltage processor has a feedback input and a feedback output, the feedback input coupled to the output voltage terminal. The control message generator has a control input and a control message output, the control input coupled to the output voltage terminal. The status signal generator has first and second signal inputs and a status signal output, the first signal input coupled to the feedback output, and the second signal input coupled to the control message output. The communication channel device has a channel input and a channel output, the channel input coupled to the status signal output, and the channel output coupled to the status signal input.

(6) A method comprises: receiving a status signal via a communication channel between primary side and secondary side circuits of a power converter, in which the status signal includes a feedback indicator and a control message, in which the feedback indicator represents a relationship between a target voltage and an output voltage of the secondary side circuit, and the control message indicates whether the output voltage is within a target voltage range. The method further comprises: determining a state of the feedback indicator, and responsive to the state of the feedback indicator, set an on-time period and an off-time period of a power stage of the primary side circuit. The method further comprises responsive to the control message, set a state of an output voltage status terminal indicating whether the output voltage is within the target voltage range.

(7) A method comprises: receiving a first voltage from an output of a power converter, and generating a feedback indicator representing a relationship between the first voltage and a target voltage. The method further comprises generating a control message indicating whether the first voltage is within a target voltage range. The method further comprises: generating a status signal including the feedback indicator and the control message, and transmitting the status signal via a communication channel between primary and secondary side circuits of the power converter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an example power converter.

(2) FIG. 2 is a schematic diagram that illustrates example internal components of the power converter of FIG. 1.

(3) FIG. 3 are waveform diagrams that illustrate example operations of the power converter of FIG. 1.

(4) FIG. 4 is a schematic diagram of an example power converter having a communication channel.

(5) FIG. 5 is a schematic diagram of example implementations of communication channel of FIG. 4.

(6) FIG. 6, FIG. 7, and FIG. 8 are schematic diagrams of example internal components of a secondary side circuit of the power converter of FIG. 4.

(7) FIGS. 9 through 12 are waveform diagrams that illustrate example signals transmitted by the secondary side circuit of FIGS. 6-8.

(8) FIG. 13 is a schematic diagram of example internal components of the example secondary side circuit of FIGS. 6-8.

(9) FIG. 14 are waveform diagrams that illustrate example signals provided by the example secondary side circuit of FIGS. 6-8.

(10) FIG. 15 is a schematic diagram of example internal components of a primary side circuit of the example power converter of FIG. 4.

(11) FIGS. 16, 17, and 18 are schematic diagrams of an example power converter that support soft-start operations.

(12) FIG. 19 is a flowchart of example soft-start operations by the power converter of FIGS. 16-18.

(13) FIG. 20 are waveform diagrams that illustrate the example soft-start operations of FIG. 19.

(14) FIG. 21A and FIG. 21B are flowcharts of an example method of operating a power converter by a primary side circuit.

(15) FIG. 22 is a flowchart of an example method of operating a power converter by a secondary side circuit.

(16) FIG. 23 is a schematic diagram of an example power converter with different semiconductor dies.

DETAILED DESCRIPTION

(17) FIG. 1 is a schematic diagram of an example power converter 100. Power converter 100 can

transfer power from a power source **102** to a load **104**. Load **104** can include electronic components that receive power from power source **102** via power converter **100**. For example, load **104** can include driver circuits for a motor. Power converter **100** can also provide a voltage $V_{sub.out}$ to load **104** based on a voltage $V_{sub.in}$ received from power source **102** based on a conversion ratio. In a case where power converter **100** is a step-up converter (e.g., a boost converter), the conversion ratio can be higher than one, and $V_{sub.out}$ can be higher than $V_{sub.in}$. In a case where power converter **100** is a step-down converter (e.g., a buck converter), the conversion ratio can be lower than one, and $V_{sub.out}$ can be lower than $V_{sub.in}$.

(18) In some examples, power converter **100** can be an isolated power convertor and includes a power stage **112** and a transformer **114**. Power stage **112** have terminals **112a** and **112b** coupled to power source **102**, and terminals **112c** and **112d** coupled to a primary side coil **114a** of transformer **114**, and secondary side coil **114b** can be coupled to load **104**. Transformer **114** can provide a galvanic isolation between power source **102** and load **104** and prevent a current from directly flowing between them, which can improve safety, while allowing transfer of power via magnetic coupling between primary side coil **114a** and secondary side coil **114b**.

(19) In some examples, isolated power converter **100** may include a rectifier **116** and a holding capacitor **120**. Rectifier **116** have terminals **116a** and **116b** coupled to secondary side coil **114b** and terminals **116c** and **116d** coupled to load **104**. Rectifier **116** can perform a rectification operation on the output of secondary side coil **114b** to maintain the relative polarities of terminals **116c** and **116d**, so that $V_{sub.out}$ can be a direct current (DC) voltage. Holding capacitor **120** can further reduce the ripple of $V_{sub.out}$ when power stage **112** is enabled to transfer power from power source **102** to load **104**. Also, when power stage **112** is disabled and load **104** is disconnected from power source **102**, holding capacitor **120** can supply a current to load **104**, and provide the $V_{sub.out}$ voltage across load **104**.

(20) FIG. 2 is a schematic diagram that illustrates example internal components of power stage **112** and rectifier **116**. Referring to FIG. 2, power stage **112** may include switches **202**, **204**, **206**, and **208**. Switches **202** and **206** can be coupled in series between terminals **112a** and **112b**, and both switches can have a terminal coupled to terminal **112c**. Also, switches **204** and **208** can be coupled in series between terminals **112a** and **112b**, and both switches can have a terminal coupled to terminal **112d**. Each switch can include a transistor, such as an n-channel field effect transistor (NFET), or a gallium nitride (GaN) high electron mobility transistor (HEMT). Switches **202**, **204**, **206**, and **208** can be controlled by, respectively, control signals **212**, **214**, **216**, and **218** (also labelled $V_{sub.G1}$, $V_{sub.G2}$, $V_{sub.G3}$, and $V_{sub.G4}$ in FIG. 2) provided by a controller **220**. Also, primary side coil **114a** and secondary side coil **114b** can be in a dot-reversed configuration, so that primary side current signal **222** (also labelled $I_{sub.P}$ in FIG. 2) can have a 180° phase shift with respect to secondary side current signal **224** (also labelled $I_{sub.S}$ in FIG. 2), as represented by the opposite arrow directions of the current signals. Further, rectifier **116** can include a rectifier bridge including diodes **230**, **232**, **234**, and **236**, with the bridge inputs coupled to terminals **116a** and **116b** and the bridge outputs coupled to terminals **116c** and **116d**. Rectifier **116** can provide a rectifier current **226** (labelled $I_{sub.R}$ in FIG. 2) to load **104** and holding capacitor **120** responsive to secondary side current signal $I_{sub.S}$.

(21) In some examples, controller **220** generates control signals $V_{sub.G1}$, $V_{sub.G2}$, $V_{sub.G3}$, and $V_{sub.G4}$ as multi-cycle signals to enable/disable, respectively, switches **202**, **204**, **206**, and **208**. During an on-time period of power stage **112**, controller **220** alternates between enabling switches **202** and **204** and enabling switches **206** and **208**, to conduct primary side current signal **222** through primary side coil **104a**. Primary side current signal **222** can induce secondary side current signal **224** at secondary side coil **114b**, and rectifier **116** can provide secondary side current signal **224** to load **104** and to charge holding capacitor **120**. Also, during the off-time period of power stage **112**, controller **220** de-asserts $V_{G.sub.1}$ and $V_{G.sub.2}$ and asserts $V_{G.sub.3}$ and $V_{G.sub.4}$. Primary side coil **114a** can be disconnected from power source **102** to remove energy stored in

transformer **114**, and holding capacitor **120** can discharge to supply a current to load **104**, and provide the $V_{sub.out}$ voltage across load **104**. By controlling the on-time period and off-time period of each cycle, controller **220** can set a particular conversion ratio between $V_{sub.out}$ and $V_{sub.in}$.

(22) FIG. 3 includes waveform graphs that illustrate example operations of power converter **100** and controller **220**. FIG. 3 includes graphs **302**, **304**, and **306**. Graph **302** illustrates an example variation of control signals $VG_{sub.1}$, $VG_{sub.2}$, $VG_{sub.3}$, and $VG_{sub.4}$ with respect to time. Graph **304** illustrates an example variation of rectifier current $I_{sub.R}$ with respect to time, and graph **306** illustrates an example variation of $V_{sub.out}$ voltage across load **104** with respect to time.

(23) Referring to graph **302**, during an on-time period of power stage **112** between times $T_{sub.0}$ and $T_{sub.1}$, controller **220** generates control signals $VG_{sub.1}$, $VG_{sub.2}$, $VG_{sub.3}$, and $VG_{sub.4}$ to alternate between enabling switches **202** and **204** and enabling switches **206** and **208** between consecutive switching cycles. In the example of FIG. 3, a first switching cycle can be between times $T_{sub.0}$ and $T_{sub.b}$, a second switching cycle can be between times $T_{sub.b}$ and $T_{sub.d}$, a third switching cycle can be between times $T_{sub.d}$ and $T_{sub.f}$, and a fourth switching cycle can be between times $T_{sub.f}$ and $T_{sub.h}$. The initial $V_{sub.out}$ voltage can be equal to $V_{sub.n}$.

(24) In the first switching cycle between $T_{sub.0}$ and $T_{sub.a}$, controller **220** asserts $VG_{sub.1}$ and $VG_{sub.2}$, as represented by pulse **310**, and de-asserts $VG_{sub.3}$ and $VG_{sub.4}$. Referring again to FIG. 2, with switches **202** and **204** enabled and switches **206** and **208** disabled, primary side current $I_{sub.P}$ can flow from power source **102** through terminal **112a**, terminal **112c**, primary side coil **114a**, terminal **112d**, and return back to power source **102** via terminal **112b**. Referring to graph **304**, between $T_{sub.0}$ and $T_{sub.a}$, primary side current $I_{sub.P}$ can also increase with time as transformer **114** stores the magnetic energy. Secondary side current $I_{sub.S}$ can be induced on the secondary side coil **114b**, which can increase with time. Also, diodes **230** and **236** can conduct secondary side current $I_{sub.S}$ as rectifier current $I_{sub.R}$, which also increases with time. Also, between $T_{sub.a}$ and $T_{sub.b}$ controller **220** de-asserts $VG_{sub.1}$ through $VG_{sub.4}$, and power source **102** can be disconnected from primary side coil **114a** and transformer **114**. Transformer **114** can discharge the stored magnetic energy to provide secondary side current $I_{sub.S}$, which can decrease with time as the magnetic energy is discharged. Rectifier current $I_{sub.R}$ also decreases with time. Within the first switching cycle (between $T_{sub.0}$ and $T_{sub.b}$) a non-zero rectifier current $I_{sub.R}$ is supplied to charge holding capacitor **120**. Accordingly, the load voltage $V_{sub.out}$ increases from the initial voltage $V_{sub.n}$ with time within the first switching cycle.

(25) In the second switching cycle between $T_{sub.b}$ and $T_{sub.d}$, controller **220** asserts $VG_{sub.3}$ and $VG_{sub.4}$, as represented by pulse **312**, and de-asserts $VG_{sub.1}$ and $VG_{sub.2}$. Referring again to FIG. 2, with switches **206** and **208** enabled and switches **202** and **204** disabled, primary side current $I_{sub.P}$ can flow from power source **102** through terminal **112a**, terminal **112d**, primary side coil **114a**, terminal **112c**, and return back to power source **102** via terminal **112b**. Both primary side current $I_{sub.P}$ and secondary side current $I_{sub.S}$ increase with time as transformer **114** stores the magnetic energy. Secondary side current $I_{sub.S}$ can flow in an opposite direction as compared with the first switching cycle, and diodes **234** and **232** can conduct secondary side current $I_{sub.S}$ as rectifier current $I_{sub.R}$ to maintain the flow direction of $I_{sub.R}$ as in the first switching cycle. Between $T_{sub.c}$ and $T_{sub.d}$ controller **220** de-asserts $VG_{sub.1}$ through $VG_{sub.4}$, power source **102** can be disconnected from primary side coil **114a** and transformer **114**, and secondary side current $I_{sub.S}$ and rectifier current $I_{sub.R}$ can decrease with time. Within the second switching cycle (between $T_{sub.c}$ and $T_{sub.d}$) a non-zero rectifier current $I_{sub.R}$ is supplied to charge holding capacitor **120**. Accordingly, the load voltage $V_{sub.out}$ continues increasing with time within the second switching cycle.

(26) For the rest of the on-time period, controller **220** continues alternating between enabling switches **202** and **204** in the third switching cycle (by asserting $VG_{sub.1}$ and $VG_{sub.2}$ and de-

asserting VG.sub.3 and VG.sub.4, represented by pulse **314**), and enabling switches **206** and **208** in the fourth switching cycle (by asserting VG.sub.3 and VG.sub.4 and de-asserting VG.sub.1 and VG.sub.2, represented by pulse **316**). Because a non-zero rectifier current I.sub.R is supplied to charge holding capacitor **120**, the load voltage V.sub.out increases with time within the on-time period, and reaches a peak voltage V.sub.p at T.sub.1 when the on-time period ends.

(27) Also, within the off-time period between T.sub.1 and T.sub.2, controller **220** disables switches **202** and **208**, and enable switches **204** and **206**. Controller **220** de-asserts VG.sub.1 and VG.sub.3 and asserts VG.sub.2 and VG.sub.4, as represented by pulse **318**. Transformer **114** can be disconnected from power source **102** to remove the stored magnetic energy, and no primary side current I.sub.P flows during the off-time period. The secondary side current I.sub.S and the rectifier current I.sub.R are also zero during the off-time period, and holding capacitor **120** can discharge to provide a current to load **104**. Accordingly, the V.sub.out voltage drops from V.sub.p back to V.sub.n.

(28) In FIG. **3**, a sum of the on-time period and the off-time period can represent a switching cycle of power stage **112**. The total duration of the on-time period (which can include multiple switching cycles of control signals VG.sub.1 through VG.sub.4) and the off-time period (in which VG.sub.1 and VG.sub.3 can be equal to T, the duration of the on-time period can be represented by DT where D is a duty cycle of the switching cycle, and the duration of the off-time period can be represented by (1-D)T. The duty cycle D can set the average V.sub.out voltage, and the conversion ratio between V.sub.out and V.sub.in.

(29) As described above, power converter **100** of FIG. **2** can provide galvanic isolation between power source **102** and load **104**. Also, controller **220** can set the conversion ratio between V.sub.out and V.sub.in by setting the duty cycle to provide a specific voltage to load **104**. However, various issues can affect the performance of power converter **100** and the safety and reliability of load **104**. Specifically, power converter **100** can introduce errors in the conversion ratio, such that the actual conversion ratio is different from a target/desired conversion ratio. Various factors can contribute to the errors, such as temperature, duty cycle error, and mismatches in the inductance between the primary and secondary side coils of transformer **114**. If controller **220** were to set the conversion ratio in an open loop operation without information of the actual V.sub.out or the actual conversion ratio, controller **220** cannot correct for the errors, and may provide an imprecise V.sub.out to load **104**.

(30) Also, if controller **220** controls power stage **112** to transfer power to load **104** via transformer **114** without information about an operation status of secondary side coil **114b**, rectifier **116**, and load **104**, various safety and reliability issues may arise. For example, a circuit failure (e.g., short circuit) may occur in power converter **100** and and/or load **104**, which may prevent secondary side coil **114b** from providing the target V.sub.out, or cause a high current and/or a high temperature in load **104** and/or power converter **100**. Without information about the operation status, controller **220** may be unable to detect such a failure and continue to enable power stage **112** to transfer power to load **104**. Such arrangements may lead to further increase in the current and/or temperature in load **104**, which can create safety hazard and damage load **104** and/or power converter **100**. Also, secondary side coil **114b** may provide an excessive V.sub.out to load **104** that can subject the circuit devices (e.g., gate drivers) of load **104** to substantial voltage stress. If controller **220** cannot detect the excessive V.sub.out and continues to enable power stage **112**, the substantial voltage stress can shorten the life time of the devices of load **104** and degrade their reliability.

(31) Further, after power converter **100** starts receiving power from power source **102** and transferring power to load **104**, it may incur considerable delay in increasing V.sub.out to a target voltage, especially for a high target voltage. If the circuit devices of load **104** are enabled or allowed to operate before V.sub.out reaches the target voltage, the electrical properties of those circuit devices may become unpredictable, which can compromise the functionality of load **104**.

(32) FIG. 4 is a schematic diagram of an example power converter **400** that can address at least some of the issues described above. In some examples, power converter **400** includes transformer **114**, a primary side circuit **402** coupled to primary side coil **114a** of transformer **114**, a secondary side **404**, and a secondary side circuit **404** coupled to secondary side coil **114b** of transformer **114**. Primary side circuit **402** includes power stage **112**, a controller **412**, and a status signal receiver **414**. Primary side circuit **402** can include an output voltage status terminal **416** (also labelled “PG” in FIG. 4) and an enable terminal **417** (labelled “EN” in FIG. 4) interfacing a host. Through output voltage status terminal **416**, controller **412** can provide a power good signal **418** to the host. Also, through enable terminal **417**, controller **412** can receive an enable signal **419** from the host. As to be described below, through power good signal **418** and enable signal **419**, controller **412** can interact with the host to perform soft-start operations to allow staged increase of $V_{sub.out}$, and to support a power gating operation to prevent load **104** from drawing power from converter **400** before $V_{sub.out}$ is within a target voltage range.

(33) In some examples, secondary side circuit **404** includes rectifier **116**, a feedback voltage processor **422**, a control message generator **424**, and a status signal generator **426**. In some examples, feedback voltage processor **422**, control message generator **424**, and status signal generator **426** can receive supply voltages from secondary side coil **114b**. Secondary side circuit **404** can also be coupled to holding capacitor **120**, and load **104** via a power gate **430**. Secondary side circuit **404** can also be coupled to a voltage divider network **432** (e.g., a resistive divider, or a capacitive divider) which can provide a feedback voltage **434** as a scaled down version of $V_{sub.out}$.

(34) Further, an output of status signal generator **426** is coupled to an input of status signal receiver **414** via a communication channel **440**. Communication channel **440** can include a capacitive channel, a magnetic channel, or an optical channel. Communication channel **440** can be implemented by a device (e.g., a capacitor, a transformer, or an optical device) that can provide electrical isolation between primary side circuit **402** and secondary side circuit **404**. FIG. 5 is a schematic diagram of example implementations of communication channel **440** between status signal generator **426** and status signal receiver **414**. Referring to FIG. 5, communication channel **440** can be implemented by a device such as a capacitor **502**, a transformer **504**, or an optical waveguide **506** (e.g., fiber optics).

(35) In some examples, primary side circuit **402**, secondary side circuit **404**, transformer **114**, and the device providing communication channel **440** (e.g., capacitor **502**, transformer **504**, or optical waveguide **506**) can each be implemented on a different semiconductor die, as shown in FIG. 23 and discussed below. The dies can be coupled by chip-level interconnects within a chip package. In some examples, each of primary side circuit **402**, secondary side circuit **404**, transformer **114**, and the device of communication channel **440** can each be a standalone component, and can be coupled by traces of a printed circuit board (PCB).

(36) Referring again to FIG. 4, from secondary side circuit **404**, controller **412** can receive a status signal **442** via status signal receiver **414** and communication channel **440**. Status signal **442** can include a feedback indicator **450** generated by feedback voltage processor **422** based on feedback voltage **434**. In some examples, feedback indicator **450** can be a hysteretic control signal that indicates whether feedback voltage **434** exceeds a target voltage. Controller **412** can extend the on-time period of a switching cycle of power stage **112** responsive to feedback indicator **450** indicating that feedback voltage **434** is below the target voltage, and start the off-time period responsive to feedback indicator **450** indicating that feedback voltage **434** is above the target voltage. In some examples, feedback indicator **450** can include a code that represents a difference/error between feedback voltage **434** and the target voltage, and controller **412** can determine the durations of the on-time and off-time periods based on the code included in feedback indicator **450**. In both examples, controller **412** generates control signals $VG_{sub.1}$ through $VG_{sub.4}$ during the respective on-time and off-time periods, to correct for errors in the conversion ratio in a closed loop

operation and to provide a more precise V.sub.out.

(37) Also, status signal **442** can include a control message **452** generated by control message generator **424**. Control message **452** can include information about an operation status of power converter **400** (or secondary side circuit **404**). The operation status can indicate, for example, whether V.sub.out is within a target voltage range that allow safe and/or proper operation of load **104**. Control message **452** can represent different types of control messages. For example, control message **452** can include a power good message, a power bad message, or a fault message. The power good message can indicate that V.sub.out is within the target voltage range. The power bad message can indicate that V.sub.out is outside the target voltage range. The fault message can indicate that for a threshold duration of time, a temperature at secondary side circuit **404** or load **104** exceeds a temperature threshold and/or V.sub.out is outside the target voltage range. As to be described below, status signal generator **426** can also transmit one of a power bad message, a power good message, or a fault message between pre-determined intervals to provide a heartbeat signal to controller **412** to indicate that secondary side circuit **404** is active and operational.

(38) Controller **412** can perform different operations in response to the different types of control messages represented by control message **452**. For example, responsive to receiving the power good message, controller **412** can set power good signal **418** to a first state (e.g., an asserted state) to indicate that V.sub.out is within the target voltage range. Controller **412** can also set power good signal **418** to a second state (e.g., a de-asserted state) responsive to receiving the power bad message. The host can enable power gate **430** to allow load **104** to receive power from power converter **400** if power good signal **418** is in the first state. The host can also disable power gate **430** to prevent load **104** from receiving power from power converter **400** if power good signal **418** is in the second state.

(39) In some examples, load **104** may be directly coupled between terminals **116c** and **116d** and not through power gate **430**. In such examples, load **104** can include circuit blocks that can receive a control signal from the host to enable or disable the circuit blocks. The host can generate the control signal to enable the circuit blocks of load **104** responsive to power good signal **418** being in the first state. The host can also generate the control signal to disable the circuit blocks of load **104** responsive to power good signal **418** being in the second state.

(40) With such arrangements, the host can allow load **104** to operate only when V.sub.out reaches the target voltage. Accordingly, the electrical properties and the functionality of load **104** can become more predictable. Also, because power gate **430** can be disabled before V.sub.out reaches the target voltage, power converter **400** needs not ramp up both V.sub.out and the current supplied to load **104** at the same time, and switches **202-208** of power stage **112** can have reduced driving capabilities. Accordingly, the sizes of switches **202-208** can be shrunk, which can reduce the footprints of power stage **112** and of power converter **400**. Also, the parasitic capacitances of the switches can be reduced, which can improve the bandwidth and efficiency of power converter **400**.

(41) In some examples, controller **412** may disable power stage **112** (e.g., by setting control signals VG.sub.1 through VG.sub.4 in a de-asserted state) and stop the power transfer to load **104** responsive to a potential fault condition in secondary side circuit **404** and/or load **104**, to improve safety and reliability. For example, controller **412** may also disable/shut down power stage **112** responsive to receiving the fault message, which can indicate that a temperature at secondary side circuit **404** (or power converter **400** as a whole) exceeds a temperature threshold for a threshold duration of time, and/or V.sub.out is outside the target voltage range for the threshold duration. In some examples, controller **412** may start a timer responsive to receiving an active enable signal **419** via enable terminal **417**. Controller **412** may disable/shut down power stage **112** to stop the transfer of power to load **104** if controller **412** does not receive the power good message before the timer expires. In some examples, controller **412** may also monitor for state changes in status signal **442**, which can indicate that secondary side circuit **404** is active. If controller **412** detects no state change in status signal **442** for a threshold duration of time, which can represent a heartbeat

interval, controller **412** may disable power stage **112**. In all these examples, controller **412** may determine that secondary side circuit **404** is likely to be in a fault condition and stop the power transfer to load **104**, which can reduce safety hazard and risk of damage to load **104**.

(42) FIG. **6**, FIG. **7**, and FIG. **8** are schematic diagrams of example internal components of secondary side circuit **404**. FIG. **6** is a schematic diagram of an example feedback voltage processor **422**. Referring to FIG. **6**, feedback voltage processor **422** can include a target voltage generator **602** and a processing circuit **604**. Target voltage generator **602** can include a voltage source to generate a target voltage **606**. As to be described below, target voltage generator **602** can include a programmable voltage source (e.g., a digital-to-analog converter (DAC)) that can be programmed to generate a variable target voltage **606** to support a soft start operation in power converter **400**.

(43) Also, processing circuit **604** can generate feedback indicator **450** based on a comparison between target voltage **606** and feedback voltage **434**, which can be a scaled version of $V_{sub.out}$. In a case where feedback indicator **450** is to be provided to controller **412** as a hysteretic control signal, processing circuit **604** can include a comparator, and the state of feedback indicator **450** can indicate whether feedback voltage **434** exceeds target voltage **606**. In a case where feedback indicator **450** represents an error/difference between feedback voltage **434** and target voltage **606**, processing circuit **604** can include an error amplifier to determine a difference between feedback voltage **434** and target voltage **606**, and generate a feedback error signal having an amplitude that represents the difference. Processing circuit **604** can also include an analog-to-digital converter (ADC) to convert the difference into a code that represents the feedback error signal, and provide the code as feedback indicator **450**.

(44) FIG. **7** is a schematic diagram of an example control message generator **424**. Referring to FIG. **7**, control message generator **424** can include reference voltage generators **702** and **704**, a temperature reference generator **706**, and a temperature sensor **708**. Control message generator **424** can also include comparators **712**, **714**, and **716**, a processing circuit **720**, and a timing circuit **722**. Reference voltage generators **702** and **704** can generate respective reference voltages **724** and **726** to be compared against feedback voltage **434**, where reference voltages **724** and **726** can represent a target voltage range for $V_{sub.out}$. Comparator **712** can compare feedback voltage **434** with reference voltage **724** to generate a decision signal **732**, and comparator **714** can compare feedback voltage **434** with reference voltage **726** to generate a decision signal **734**. Based on the states of decision signals **732** and **734**, processing circuit **720** can determine whether $V_{sub.out}$ is within the target voltage range. If $V_{sub.out}$ is within the target voltage range, processing circuit **720** can provide control message **452** including the power good message, which allows the host to enable power gate **430** and/or circuit blocks in load **104**.

(45) Also, if $V_{sub.out}$ is outside the target voltage range, processing circuit **720** can determine the duration in which $V_{sub.out}$ is outside the target voltage range, based on a timing signal from timing circuit **722**. Before a threshold duration of time is reached, processing circuit **720** can provide control message **452** including the power bad message. If the threshold duration of time is reached and $V_{sub.out}$ remains outside the target voltage range, processing circuit **720** can provide control message **452** including the fault message, which can cause controller **412** to disable power stage **112**.

(46) Further, comparator **716** can compare a threshold temperature signal **736** provided by threshold temperature generator **706** with a temperature measurement signal **738** provided by temperature sensor **708** to generate a decision signal **746**. Temperature measurement signal **738** can represent a temperature of secondary side circuit **404** and/or load **104**, and both threshold temperature signal **736** and temperature measurement signal **738** can be in the form of voltages. If a state of decision signal **746** indicates that the temperature exceeds the temperature threshold, and decision signal **746** remains in that state over the threshold duration of time, processing circuit **720** can also provide control message **452** including the fault message, which can cause controller **412**

to disable power stage **112** as described above.

(47) FIG. **8** is a schematic diagram of an example status signal generator **426**. Referring to FIG. **8**, status signal generator **426** can include a signal combination circuit **802**, a timing circuit **804**, and a transmitter **806**. Signal combination circuit **802** can receive feedback indicator **450** (from feedback voltage processor **422**) and control message **452** (from control message generator **424**), and generate a data signal **810** including a serial stream of symbols representing feedback indicator **450** and control message **452**. Transmitter **806** can generate status signal **442** based on the logical states of the symbols of data signal **810** and transmit status signal **442** over communication channel **440**.

(48) Signal combination circuit **802** can receive a clock signal **808** from timing circuit **804** and synchronize both feedback indicator **450** and control message **452** to clock signal **808**. Signal combination circuit **802** can also generate symbols representing the synchronized feedback indicator **450** and control message **452**, and insert the symbols into data signal **810** at symbol periods defined based on clock signal **808**. For example, signal combination circuit **802** may insert symbols **812** representing feedback indicator **450** (or control message **452**) into data signal **810** within period P.sub.1, and insert symbols **814** representing control message **452** (or feedback indicator **450**) into data signal **810** within period P.sub.2.

(49) Signal combination circuit **802** can determine the order of transmission of feedback indicator **450** and control message **452** in data signal **810** based on various techniques, such as based on a predetermined schedule, or based on priority. FIG. **9** are waveform diagrams that illustrate example interleaving patterns of feedback indicator **450** and control message **452** in data signal **810**. In FIG. **9**, symbols **812** can represent feedback indicator **450**, and symbols **814** can represent control message **452**.

(50) Referring to FIG. **9**, in pattern **900**, symbols **812** and **814** can be interleaved following a predetermined schedule. At the scheduled time of transmission, signal combination circuit **802** can receive one of feedback indicator **450** or control message **452**, convert the sampled signal into symbols, and insert the symbols into data signal **810**. For example, feedback indicator **450** is to be transmitted within period P.sub.1 between T.sub.0 and T.sub.1 and within period P.sub.3 between T.sub.2 and T.sub.3, and control message **452** is to be transmitted within period P.sub.2 between T.sub.1 and T.sub.2 and within period P.sub.4 between T.sub.3 and T.sub.4. Accordingly, within periods P.sub.1 and P.sub.3, signal combination circuit **802** can receive feedback indicator **450**, generate respective symbols **812a** and **812b** based on the state of the feedback indicator, and insert symbols **812a** and **812b** into data signal **810** within periods P.sub.1 and P.sub.3. Also, within periods P.sub.2 and P.sub.4, signal combination circuit **802** can receive control message **452**, generate respective symbols **814a** and **814b** based on the state of the feedback indicator, and insert symbols **814a** and **814b** into data signal **810** within periods P.sub.2 and P.sub.4.

(51) In some examples, signal combination circuit **802** can determine the transmission order of symbols **812** and **814** based on priority, where signal combination circuit **802** can interrupt the transmission of a lower priority symbol in data signal **810** in order to transmit the higher priority symbol. For example, referring to pattern **902**, signal combination circuit **802** may interrupt the transmission of symbols **812a** at T.sub.1 to insert symbols **814** into data signal **810**, based on symbols **814** (representing control message **452**) having a higher priority than symbols **812a** (representing feedback indicator **450**). After transmission of symbols **814**, signal combination circuit **802** can insert symbols **812b** into data signal **810**.

(52) Signal combination circuit **802** may determine that control message **452** has a higher transmission priority than feedback indicator **450** in various operation conditions. For example, as described above, controller **412** may start a timer responsive to receiving an active enable signal **419**, and may disable/shut down power stage **112** if it does not receive the power good message before the timer expires. Accordingly, responsive to generating the first power good message after power converter **400** starts up, signal combination circuit **802** may prioritize transmission of the power good message to reduce the likelihood of controller **412** disabling power stage **112**.

Referring again to pattern **902**, signal combination circuit **802** may interrupt the transmission of symbols **812a** representing feedback indicator **450** at T.sub.1, insert symbols **814** representing the power good message in data signal **810** between T.sub.1 and T.sub.2, and then transmit symbols **812b** representing feedback indicator **450** starting at T.sub.2.

(53) In some examples, if controller **412** detects no state change in status signal **442** for a threshold duration of time, controller **412** may disable power stage **112**. Accordingly, signal combination circuit **802** can measure a duration of time in which feedback indicator **450** is in a particular state based on clock signal **808**. If the duration of time reaches or exceeds the threshold duration of time, signal combination circuit **802** can insert symbols **814** representing a message into data signal **810** as a heartbeat signal, to avoid controller **412** disabling power stage **112** due to suspected inactivity of secondary side circuit **404**. Accordingly, if feedback indicator **450** remains in the particular state, signal combination circuit **802** can insert a message into data signal **810** between intervals defined by the threshold duration. Referring to pattern **902**, signal combination circuit **802** may interrupt the transmission of symbols **812a** representing feedback indicator **450** at T.sub.1 due to feedback indicator **450** staying at a particular state for the threshold duration of time, and insert symbols **814** representing a message in data signal **810** between T.sub.1 and T.sub.2. Signal combination circuit **802** can determine a state of control message **452** when the threshold duration is reached, and insert one of a power good message, a power bad message, or a fault message as the heartbeat signal based on control message **452**.

(54) In some examples, signal combination circuit **802** can also generate one or more symbols for status signal **442** representing a combination of feedback indicator **450** and control message **452**. For example, both signal combination circuit **802** and status signal receiver **414** can store a mapping table that maps between different symbols and different combinations of states of feedback indicator **450** and types of control message **452**. Responsive to receiving feedback indicator **450** and control message **452**, signal combination circuit **802** can determine a state of feedback indicator **450** and a type of control message **452**. From the mapping table, signal combination circuit **802** can determine one or more symbols mapped to the particular state of feedback indicator **450** and the particular type of control message **452**, and include the one or more symbols in data signal **810**. Also, upon receiving status signal **442** including the one or more symbols, status signal receiver **414** can decode the symbols using the mapping table to obtain the state of feedback indicator **450** and the type of control message **452** represented by the symbols.

(55) In some examples, signal combination circuit **802** can insert one or more pulses in data signal **810** as symbols to represent feedback indicator **450** and control message **452**. The pulses of feedback indicator **450** can represent, for example, a result of a comparison between feedback voltage **434** and target voltage **606**, or a code representing an error/difference between feedback voltage **434** and target voltage **606**. Also, different types of control messages **452** (e.g., power good message, power bad message, and fault message) can be represented by different number of pulses/symbols. The pulses of control messages **452** can have a much higher bandwidth than feedback indicator **450**, and adjacent state transitions in control message **452** are separated by a much shorter time period than adjacent state transitions in feedback indicator **450**. Also, signal combination circuit **802** can set the voltage state of the pulses to be opposite to the voltage state of symbols of feedback indicator **450** preceding the symbols of control message **452**. Such arrangements can simplify the extraction of feedback indicator **450** and control message **452** from status signal **442**, where controller **412** can apply a low pass filter to filter out the pulses in data signal **810**, and extract feedback indicator **450** from the filtered data signal.

(56) FIG. **10**, FIG. **11**, and FIG. **12** are waveform diagrams of example data signal **810** generated by signal combination circuit **802**. In FIGS. **10** through **12**, data signal **810** can include symbols **812a**, **812b**, and **814**, where symbols **812** can represent a hysteretic control signal. Data signal **810** can have one of two voltage states V.sub.L and V.sub.H each representing different logical states, where V.sub.L can represent a logical zero or a de-asserted state, and V.sub.H can represent a

logical one or an asserted state. Symbols **812a** and **812b** can span respectively periods of P.sub.1 and P.sub.3, and symbols **814** can span a period of P.sub.2.

(57) FIG. **10** illustrates waveform graphs **1000** and **1002** of example symbols **812a** and **812b** representing feedback indicator **450** and example symbols **814** representing one type of control message **452**, such as a power good message. In FIG. **10**, symbols **814** can include a symbol including a single pulse having a pulse width T.sub.pulse, followed by an end symbol having a width of T.sub.end, which can have a constant relationship with T.sub.pulse across different kinds of control messages. The pulse width T.sub.pulse of symbols **814** can be smaller than the minimum time between state transitions in symbols **812** (of feedback indicator **450**), to allow controller **412** to distinguish between feedback indicator **450** and control message **452**.

(58) FIG. **11** illustrates waveform graphs **1100** and **1102** of example symbols **814** representing another type of control message **452**, such as a power bad message. In FIG. **11**, symbols **814** can include a symbol having two pulses each having a pulse width T.sub.pulse, followed by an end symbol having a width of T.sub.end. Further, FIG. **12** illustrates waveform graphs **1200** and **1202** of example symbols **814** representing another type of control message **452**, such as a fault message. In FIG. **12**, symbols **814** can include a symbol including three pulses each having a pulse width T.sub.pulse, followed by an end symbol having a width of T.sub.end. In some examples, the duration of T.sub.end can be a multiple of the duration of T.sub.pulse. For example, T.sub.pulse can span three cycles of clock signal **808**, and T.sub.end can span six clock cycles of clock signal **808**.

(59) In FIGS. **11** and **12**, adjacent pulses are also separated by a time period T.sub.pulse, in which data signal **810** has an opposite voltage state from the pulses. Also, the pulses of symbols **814** can have opposite voltage states to symbols **812a** and **812b**. For example, in graphs **1000**, **1100**, and **1200**, symbols **812a** and **812b** can be in the V.sub.H state and symbols **814** can be in the V.sub.L state. Also, in graphs **1002**, **1102**, and **1202**, symbols **812a** and **812b** can be in the V.sub.L state and symbols **814** can be in the V.sub.H state.

(60) FIG. **13** is a schematic diagram of example internal components of transmitter **806**, and FIG. **14** include waveform diagrams of example data signal **810** and status signal **442**. Referring to FIG. **13**, transmitter **806** can include a signal source **1302** and a modulator **1304**. Modulator **1304** can receive data signal **810** including the symbols for feedback indicator **450** and control message **452** from signal combination circuit **802**. Modulator **1304** can also receive a signal **1310** from signal source **1302**. Modulator **1304** can modulate signal **1310** based on the voltage states of the symbols, and provide the modulated signal as status signal **442**.

(61) Signal source **1302** can provide different types of signal **1310** depending on the properties of communication channel **440**. For example, in a case where communication channel **440** includes a capacitive channel provided by capacitor **502** of FIG. **5**, capacitor **502** can be configured as a high pass filter. Signal source **1302** can provide an oscillating carrier signal **1310** having a higher switching frequency than data signal **810**, and modulator **1304** can perform an on-off keying modulation operation on carrier signal **1310**. Because capacitor **502** is configured as a high pass filter, it can transmit a high frequency carrier signal **1310** with reduced or no loss.

(62) FIG. **14** includes waveform diagrams **1400** and **1402**. Waveform diagram **1400** illustrates an example data signal **810**, and waveform diagram **1402** illustrates an example status signal **442** generated by modulator **1304** responsive to the example data signal **810**. Referring to FIG. **14**, between times T.sub.0 and T.sub.1 and between times T.sub.2 and T.sub.3 when data signal **810** is in the V.sub.H state and represents one or more logical one symbols, modulator **1304** can allow carrier signal **1310** to pass through as status signal **442**. Also, when data signal **810** is in the V.sub.L state and represents one or more logical zero symbols between T.sub.2 and T.sub.3, modulator **1304** can block out carrier signal **1310** from status signal **442**. Accordingly, signal status receiver **414** of primary side circuit **402** can recover the symbols from status signal **442** by detecting the presence or absence of carrier signal **1310** in status signal **442**, where the presence of carrier signal **1310** can indicate a logical one and the absence of carrier signal **1310** can indicate a logical zero.

(63) Referring again to FIG. 13, signal source **1302** can provide other types of signal **1310** for other types of communication channel **440**. For example, in a case where communication channel **440** includes an optical channel (e.g., provided by optical waveguide **506** of FIG. 5), signal source **1302** can include a light source (e.g., a laser source) to provide a light signal **1310**. Modulator **1304** can transmit or block light signal **1310** based on the state of data signal **810**, and signal status receiver **414** of primary side circuit **402** can recover the symbols from status signal **442** by detecting the presence or absence of light signal **1310** in status signal **442**. Also, in a case where communication channel **440** includes a magnetic channel (e.g., provide by transformer **504** of FIG. 5), signal source **1302** can include a direct current (DC) voltage source to provide a DC voltage **1310**, and modulator **1304** can modulate the DC voltage **1310** to provide status signal **442**. Signal status receiver **414** of primary side circuit **402** can recover the symbols from status signal **442** by detecting the voltage state of status signal **442**. In some examples, signal source **1302** can also provide an oscillating carrier signal **1310**, which can be modulated by modulator **1304**, for transmission of status signal **442** over the magnetic channel.

(64) FIG. 15 is a schematic diagram of example internal components of controller **412** of primary side circuit **402**. Referring to FIG. 15, controller **412** can include a signal extractor **1502**, a timing circuit **1504**, a switching cycle determination circuit **1506**, and a control signal generator **1508**. Signal extractor **1502** can receive a data signal **1510**, which can include symbols recovered from status signal **442** by signal status receiver **414**. Signal extractor **1502** can also receive a clock signal **1512** from timing circuit **1504**. Clock signal **1512** can have the same frequency as clock signal **808** of status signal generator **426** of secondary side circuit **404**.

(65) Signal extractor **1502** can extract data signal **1510** from status signal **442**. In a case where status signal **442** includes a carrier signal modulated by the symbols of feedback indicator **450** and control message **452**, signal extractor **1502** can set data signal **1510** to a first state (e.g., V.sub.H or asserted state) in a signal period where the carrier signal is detected, and set data signal **1510** to a second state (e.g., V.sub.L state) in a signal period where the carrier signal is not detected. In a case where status signal **442** are in the form of pulses, signal extractor **1502** can set the state of data signal **1510** based on the amplitude of status signal **442**. Signal extractor **1502** can also synchronize data signal **1510** with clock signal **1512**, and perform filtering operations on the synchronized data signal **1510** to remove noise. Signal extractor **1502** can also separate the symbols of feedback indicator **450** from the symbols of control message **452** by performing another low pass filtering, based on the symbols of feedback indicator **450** having a lower bandwidth than those of control messages **452**.

(66) From the extracted symbols, signal extractor **1502** can determine a state of feedback indicator **450**. For example, in a case where feedback indicator **450** includes a hysteretic control signal, signal extractor **1502** can provide feedback indicator **450** having a first state (e.g., V.sub.H state in FIGS. 10-12) for an extracted symbol representing a logical one, and provide feedback indicator **450** having a second state (e.g., V.sub.L state in FIGS. 10-12) for an extracted symbol representing a logical zero. Also, in a case where feedback indicator **450** includes a code representing an error/difference between feedback voltage **434** and target voltage **606**, signal extractor **1502** can determine the code from the symbol, and provide the code as feedback indicator **450**. Signal extractor **1502** can also search for symbols representing different types of control messages **452** by detecting a number of transitions each separated by T.sub.pulse, and by detecting the end symbol.

(67) Switching cycle determination circuit **1506** can generate a switching cycle signal **1507** that represents the switching cycles of power stage **112**. The state of switching cycle signal **1507** can define the on-time and off-time periods of each switching cycle. For example, switching cycle signal **1507** having a first state (e.g., an asserted state) can indicate that power stage **112** is to be turned on, and switching cycle signal **1507** having a second state (e.g., a de-asserted state) can indicate that power stage **112** is to be turned off. Switching cycle determination circuit **1506** can set the duty cycle of switching cycle signal **1507** (and the on-time and off-time periods durations)

based on feedback indicator **450** extracted by signal extractor **1502**. For example, if feedback indicator **450** is a hysteretic control signal that indicates whether feedback voltage **434** exceeds a target voltage, switching cycle determination circuit **1506** can extend the on-time period of a cycle of switching cycle signal **1507** responsive to feedback indicator **450** indicating that feedback voltage **434** is below the target voltage. Also, switching cycle determination circuit **1506** can start the off-time period of that cycle (or the next cycle) responsive to feedback indicator **450** indicating that feedback voltage **434** is above the target voltage. As another example, if feedback indicator **450** includes a code that represents a difference/error between feedback voltage **434** and the target voltage, switching cycle determination circuit **1506** can determine the durations of the on-time and off-time periods based on the code included in feedback indicator **450**.

(68) Also, in certain operation conditions, switching cycle determination circuit **1506** can disable power stage **112** by setting switching cycle signal **1507** in the second/de-asserted state. For example, switching cycle determination circuit **1506** can include a timer **1516** to measure a duration of time in which feedback indicator **450** remains at a particular state (e.g., in the V.sub.H or V.sub.L state). If the duration exceeds a threshold, switching cycle determination circuit **1506** can disable power stage **112** by setting switching cycle signal **1507** in the second/de-asserted state.

(69) Further, control signal generator **1508** can receive and handle control message **452** extracted by signal extractor **1502**, and can also provide control signals **212** through **218** (VG1 through VG4) based on switching cycle signal **1507** and control message **452**. For example, responsive to detecting the power good message in control message **452**, control signal generator **1508** can set power good signal **418** at output voltage status terminal **416** to a first state (e.g., an asserted state) to indicate to the host that V.sub.out is within the target voltage range. Control signal generator **1508** can also set power good signal **418** to a second state (e.g., a de-asserted state) responsive to, for example, detecting the power bad message or the fault message in control message **452**, or otherwise determining to disable power stage **112**.

(70) Also, if control message **452** does not include a fault message, control signal generator **1508** can generate control signals VG.sub.1 through VG.sub.4 responsive to the state of switching cycle signal **1507**. For example, referring again to FIG. 3, responsive to switching cycle signal **1507** being in the first state, which can indicate the on-time period of a switching cycle, control signal generator **1508** can generate control signals VG.sub.1 through VG.sub.4 as multi-cycle control signals to transfer power from power source **102** to load **104**. Also, responsive to switching cycle signal **1507** being in the second state, which can indicate the off-time period of the switching cycle (or another switching cycle), control signal generator **1508** can de-assert VG.sub.1 and VG.sub.3 and assert VG.sub.2 and VG.sub.4 to disconnect load **104** from power source **102**. In some examples, switching cycle signal **1507** can provide the duty cycle information, and control signal generator **1508** can determine the switching frequency information (e.g., based on programming from the host), and generate control signals VG.sub.1 through VG.sub.4 based on the switching frequency information and the duty cycle information.

(71) Control signal generator **1508** can also override switching cycle signal **1507** and disable power stage **112** in certain operation conditions. For example, responsive to detecting a potential fault condition, control signal generator **1508** can disable power stage **112** by de-asserting VG.sub.1 through VG.sub.4 to power stage **112**. Control signal generator **1508** can detect the potential fault condition based on, for example, detecting a fault message in control message **452**. In some examples, control signal generator **1508** can also include a timer **1518**. Control signal generator **1508** can start timer **1518** responsive to receiving an active enable signal **419** via enable terminal **417**. If control signal generator **1508** does not detect the power good message in control message **452** prior to the expiration of timer **1518**, control signal generator **1508** can disable power stage **112**.

(72) In some examples, power converter **400** can support a respective soft start operation in each of primary side circuit **402** and secondary side circuit **404**. As part of the soft start operations, power

converter **400** can increase $V_{sub.out}$ in a sequence of target voltages. Also, during the soft start operation, controller **412** can begin adjusting $V_{sub.out}$ with an open loop operation, and then switch to adjusting $V_{sub.out}$ with a closed loop operation based on feedback indicator **450** after receiving status signal **442**, which can indicate that secondary side circuit **404** has been fully powered up (e.g., by drawing power from rectifier **116**) and is operational. With such arrangements, the ramping up of $V_{sub.out}$ can be performed at a slower pace, which allows for reduced driving capabilities and reduced sizes of switches **202** through **208** of power stage **112**, and improve the bandwidth and efficiency of power converter **400**. Also, because controller **412** adjusts $V_{sub.out}$ with a closed loop operation after secondary side circuit **404** fully powers up and feedback indicator **450** becomes reliable, the precision in providing a specific $V_{sub.out}$ to load **104** can also be improved. All these can improve the performance of power converter **400**.

(73) FIG. **16** is a schematic diagram of example internal components of power converter **400** to support the soft start operations. Referring to FIG. **16**, controller **412** can include a primary side soft start control circuit **1602**, and secondary side circuit **404** can include a secondary side soft start control circuit **1604**. Primary side soft start control circuit **1602** can support a primary side soft start operation, and secondary side soft start control circuit **1604** can support a secondary side soft start operation.

(74) FIG. **17** is a schematic diagram of example internal components of controller **412** including primary side soft start control circuit **1602**. As part of the primary side soft start operation on primary side circuit **402**, primary side soft start control circuit **1602** can provide control signals to power stage **112** to increase $V_{sub.out}$ in an open loop operation before secondary side circuit **404** powers up.

(75) Specifically, primary side soft start control circuit **1602** can be coupled to enable terminal **417** and switching cycle determination circuit **1506** and can also receive data signal **1510** from status signal receiver **414**. In some examples, primary side soft start control circuit **1602** may also be coupled to control signal generator **1508**. Responsive to receiving enable signal **419** at enable terminal **417**, primary side soft start control circuit **1602** can start the open loop operation and provide an open loop control signal **1702** to switching cycle determination circuit **1506**, which can set the switching frequency and duty cycle of switching cycle signal **1507** based on open loop control signal **1702**. In some examples, control signal generator **1508** can also set the switching frequencies of control signals $VG_{sub.1}$ through $VG_{sub.4}$ based on open loop control signal **1702**, and set the states of the control signals based on the duty cycle of switching cycle signal **1507**. The control signals $VG_{sub.1}$ through $VG_{sub.4}$ can have different switching frequencies in the open loop operation versus the closed loop operation, to enable control of ramp up time of the $V_{sub.out}$ voltage.

(76) Responsive to open loop control signal **1702**, switching cycle determination circuit **1506** can provide switching cycle signal **1507** at the switching frequency and having the duty cycle specified by open loop control signal **1702**, and control signal generator **1508** can generate control signals $VG_{sub.1}$ through $VG_{sub.4}$ responsive to a state of switching cycle signal **1507**, to start a power transfer from power source **102** to secondary side circuit **404**. Primary side soft start control circuit **1602** can also update the duty cycle specified in open loop control signal **1702** based on a pre-determined sequence. For example, primary side soft start control circuit **1602** can provide open loop control signal **1702** that specify a starting duty cycle at 12%, followed by subsequent open loop control signals **1702** that specify duty cycles of 25%, 50%, and 100%.

(77) Primary side soft start control circuit **1602** can provide open loop control signal that specify a particular duty cycle across multiple switching cycles, and move to the next duty cycle if it does not receive data signal **1510**. But if primary side soft start control circuit **1602** receives data signal **1510** from status signal receiver **414**, which generates data signal **1510** by extracting symbols from status signal **442**, primary side soft start control circuit **1602** can determine that secondary side circuit **404** is powered up and is capable of transmitting status signal **442**. Accordingly, primary

side soft start control circuit **1602** can stop the open loop operation. Switching cycle determination circuit **1506** can also start the closed loop operation and adjust the duty cycle/on-time period of the control signals based on feedback indicator **450** extracted from status signal **442**.

(78) Also, referring again to FIG. **16**, secondary side soft start control circuit **1604** can be coupled to feedback voltage processor **422**, status signal generator **426**, and voltage divider network **432**. Before the secondary side soft start operation starts, secondary side soft start control circuit **1604** can disable status signal generator **426**, and monitor the ramping of feedback voltage **434** during the primary side soft start operation. Secondary side soft start control circuit **1604** can start the secondary side soft start operation if feedback voltage **434** exceeds a soft start threshold voltage, which can represent a undervoltage-lockout (UVLO) threshold for secondary side circuit **404**. During the secondary side soft start operation, secondary side soft start control circuit **1604** can increase the target voltage for generation of feedback indicator **450** in steps until a final target voltage is reached, which allows controller **412** to also increase $V_{sub.out}$ in steps responsive to feedback indicator **450** to the final target voltage.

(79) In some examples, control signal generator **1508** can operate to reduce the occurrence of a lockup event, in which after being disabled, power stage **112** is enabled when power converter **400** still provides a large residual $V_{sub.out}$ voltage. This may cause secondary side soft start control circuit **1604** to start the secondary side soft start operation if feedback voltage **434** (due to the large residual $V_{sub.out}$ voltage) exceeds the UVLO threshold, when the rest of power converter **400** are not yet properly reset.

(80) To reduce or prevent the lockup event, after control signal generator **1508** receives a fault message and provides control signals **214** through **218** ($VG_{sub.1}$ through $VG_{sub.4}$) to disable power stage **112**, control signal generator **1508** can delay the start of the primary side soft start operation when signal extractor **1502** still receives status signal **442** from secondary side circuit **404**. The reception of status signal **442** can indicate that secondary side circuit **404** is not yet powered down and there can still be a large residual $V_{sub.out}$ voltage. Control signal generator **1508** can ignore enable signal **419** and maintain power stage **112** in the disabled state until communication channel **440** becomes inactive and signal extractor **1502** no longer receives status signal **442** from secondary side circuit **404**. In some examples, control signal generator **1508** can also monitor for status signal **442** within a time window after receiving enable signal **419**, and start the primary side soft start operation if communication channel **440** remains inactive within the window. For example, control signal generator **1508** can start timer **1528** responsive to receiving enable signal **419**, and start the primary side soft start operation if no status signal **442** is received before timer **1528** expires. Control signal generator **1508** can also extend the time window (e.g., by resetting timer **1528**) responsive to receiving status signal **442** within the time window. With such arrangements, power converter **400** can delay the primary side soft start operation until the $V_{sub.out}$ voltage falls below the UVLO threshold, and power converter **400** can be properly reset, prior to starting the soft start sequence, which can reduce or prevent the occurrence of lockup event.

(81) FIG. **18** is a schematic diagram of example internal components of secondary side soft start control circuit **1604**. Referring to FIG. **18**, secondary side soft start control circuit **1604** can include a threshold voltage generator **1802**, a comparator **1806**, a status signal control circuit **1808**, and a target voltage control circuit **1810**. Threshold voltage generator **1802** can generate a threshold voltage **1812** representing an UVLO threshold. Comparator **1806** can compare feedback voltage **434** with threshold voltage **1812** to generate a decision signal **1832**, and provide decision signal **1832** to status signal control circuit **1808** and target voltage control circuit **1810**. Responsive to decision signal **1832**, status signal control circuit **1808** can generate a control signal **1842** to enable/disable status signal generator **426**, and target voltage control circuit **1810** can generate a control signal **1850** to set a target voltage for feedback voltage processor **422**.

(82) Specifically, if feedback voltage **434** exceeds threshold voltage **1812**, the supply voltages

provided by secondary side coil **114b** for feedback voltage processor **422**, control message generator **424**, and status signal generator **426** can be sufficiently high to allow these circuits to operate properly. Accordingly, if a state of decision signal **1832** indicates that feedback voltage **434** is below threshold voltage **1812**, status signal control circuit **1808** can provide control signal **1842** to disable status signal generator **426**, or otherwise to prevent status signal generator **426** from transmitting status signal **442** via communication channel **440**. But if a state of decision signal **1832** indicates that feedback voltage **434** is above threshold voltage **1812**, status signal control circuit **1808** can provide control signal **1842** to enable status signal generator **426**, or to allow status signal generator **426** to transmit status signal **442**.

(83) Also, if a state of decision signal **1832** indicates that feedback voltage **434** is above threshold voltage **1812**, target voltage control circuit **1810** can provide control signal **1850** to target voltage generator **602** to set the target voltage **606**, which processing circuit **604** of feedback voltage processor **422** can compare with feedback voltage **434** to generate feedback indicator **450**. As part of the secondary side soft start operation, target voltage control circuit **1810** can provide control signal **1850** to target voltage generator **602** to increase target voltage **606** based on a pre-determined sequence of target voltages. Processing circuit **604** can adjust feedback indicator **450** responsive to the increase in target voltage **606**, which allows controller **412** to also adjust the on-time and off-time period durations based on the adjusted feedback indicator **450**, and provide the control signals VG.sub.1 through VG.sub.4 within the respective on-time and off-time periods.

(84) Further, after target voltage control circuit **1810** provides the last target voltage in the sequence, target voltage control circuit **1810** can provide a complete signal **1852** back to status signal control circuit **1808**. Status signal control circuit **1808** can include complete signal **1852** in control signal **1842**. Responsive to receiving complete signal **1852**, status signal generator **426** can prioritize transmission of power good message in status signal **442** to controller **412**, to avoid controller **412** receiving the power good message after timer **1518** expires and disabling power stage **112**.

(85) FIG. **19** includes a flowchart **1900** of example primary side and secondary side soft start operations, and FIG. **20** includes waveform diagrams that illustrate the state of various signals of power converter **400** during the soft start operations described in FIG. **19**. In FIG. **20**, graph **2002** represents V.sub.in, graph **2004** represents enable signal **419** (labelled “EN”), graph **2006** represents signal **419** (labelled “PG”), and graph **2008** represents V.sub.out. Also, graph **2010** represents decision signal **1832** (labelled “V.sub.out>UVLO?”), and graph **2012** represents decision signal **732** (labelled “V.sub.out>UVP?”) which indicates whether V.sub.out exceeds a lower limit of the target voltage range. Also, graph **2014** represents feedback indicator **450** (labelled “FB”) that can include a hysteretic control signal, and graph **2016** represents an internal flag (labelled “Done”) that indicates whether the secondary side soft start operation completes. Further, graph **2018** represents a target voltage code (labelled “target voltage code”) included in control signal **1850** and graph **2020** represents status signal **442**. Graph **2022** represents switching cycle signal **1507**. The on-time periods can be represented by switching cycle signal **1507** being in the asserted state, and the off-time periods can be represented by switching cycle signal **1507** being in the de-asserted state.

(86) Referring to FIG. **19**, in step **1902**, primary side soft start operation begins. Primary side soft start control circuit **1602** can start the soft start operation responsive to the assertion of the EN signal (enable signal **419**) by, for example, a host that interfaces with power converter **400**.

(87) In some examples, as described above, control signal generator **1508** can start timer **1518** after providing de-asserted control signals VG.sub.1 through VG.sub.4 to disable power stage **112**, and start the soft start operation after timer **1518** expires (and EN signal is also asserted), to provide sufficient time for power converter **400** to discharge and to properly reset before enabling power stage **112**. Such arrangements can reduce the occurrence of a lockup event.

(88) In step **1904**, primary side soft start control circuit **1602** can set the duty cycle of switching

cycle signal **1507** (and the on-time period durations of power stage **112**) following a predetermined sequence. This is represented between times $T_{sub.0}$ and $T_{sub.2}$ in FIG. **20**, when switching cycle determination circuit **1506** provides switching cycle signal **1507** at a high frequency in an open loop operation, and control signal generator **1508** can provide control signals $V_{G.sub.1}$ through $V_{G.sub.4}$ also at a high frequency responsive to switching cycle signal **1507**. Also, between $T_{sub.0}$ and $T_{sub.2}$, controller **412** increases the duty cycle of switching cycle signal **1507** (and the on-time period duration of power stage **112**) based on the sequence, and $V_{sub.out}$ can increase accordingly. (89) Referring again to FIG. **19**, primary side soft start control circuit **1602** can proceed to step **1906** and determine whether status signal **442** is received. Primary side soft start control circuit **1602** can proceed back to step **1904** and continue the open loop operation if status signal **442** is not received. Referring again to FIG. **20**, if communication channel **440** remains in an inactive state and secondary side circuit **404** does not transmit status signal **442** between $T_{sub.0}$ and $T_{sub.2}$, primary side soft start control circuit **1602** can continue providing control signals $V_{G.sub.1}$ through $V_{G.sub.4}$ in the open loop operation.

(90) In step **1908**, secondary side circuit **404**, including feedback voltage processor **450**, control message generator **424**, status signal generator **426**, and secondary side soft start control circuit **1604** can wake up from a power down state. This can be due to these components receiving their supply voltages from secondary side coil **114b** and their supply voltages increase with $V_{sub.out}$.

(91) In step **1910**, secondary side soft start control circuit **1604** can determine whether $V_{sub.out}$ exceeds an UVLO threshold based on, for example, comparing feedback voltage **434** with threshold voltage **1812**. Secondary side soft start control circuit **1604** can disable status signal generator **426** and block the transmission of status signal **442** if $V_{sub.out}$ is below UVLO threshold. This is also illustrated in graph **2010** of FIG. **20**, where between $T_{sub.1}$ and $T_{sub.2}$ decision signal **1832** is in an asserted state indicating that $V_{sub.out}$ is below UVLO threshold, and communication channel **440** remains in the inactive state.

(92) Referring again to FIG. **19**, in step **1910**, if $V_{sub.out}$ exceeds the UVLO threshold, secondary side soft start control circuit **1604** can proceed to step **1912** and allow status signal generator **426** to transmit a status signal **442**. As described above, $V_{sub.out}$ exceeding the UVLO threshold can indicate that the supply voltages for feedback voltage processor **422**, control message generator **424**, and status signal generator **426** are sufficiently high to allow these circuits to operate properly. Because $V_{sub.out}$ is outside the target range, status signal generator **426** can transmit a status signal **442** including a power bad message to primary side circuit **402**, in step **1912**. Responsive to receiving status signal **442**, and if timer **1516** has not expired (e.g., due to feedback indicator **450** changing states, or a heart beat message is received), primary side soft start control circuit **1602** can end the primary side soft start operation, in step **1914**. These are also illustrated in FIG. **20**, where between $T_{sub.2}$ and $T_{sub.3}$ primary side soft start control circuit **1602** receives the power bad message (represented by two pulses having $V_{sub.L}$ state) and ends the primary side soft start operation. If timer **1516** has expired when status signal **442** is received, controller **412** may disable power stage **112**.

(93) Referring again to FIG. **19**, in step **1916**, secondary side soft start control circuit **1604** can start the secondary side soft start operation, responsive to $V_{sub.out}$ exceeding the UVLO threshold. As part of the soft start operation, secondary side soft start control circuit **1604** (and target voltage control circuit **1810**) can provide control signal **1850** to target voltage generator **602** to increase target voltage **606** based on a pre-determined sequence, in step **1918**. Processing circuit **604** can generate feedback indicator **450** responsive to target voltage **606** and feedback voltage **434** in step **1920**, and transmit feedback indicator as part of status signal **442** in step **1922**. Controller **412** can then adjust switching cycle signal **1507** (and the on-time period duration of power stage **112**) based on feedback indicator **450**, in step **1924**. Secondary side soft start control circuit **1604** can determine whether $V_{sub.out}$ is within a target voltage range based on decision signals **732** and **734** in step **1926**, and can repeat steps **1918** and **1920** if $V_{sub.out}$ is outside the target range.

(94) The operations described above can be illustrated in FIG. 20. Specifically, between T.sub.2 and T.sub.3, feedback voltage processor **422** provides feedback indicator **450** having the V.sub.H state, and status signal generator **426** can include feedback indicator **450** having the V.sub.H state before the pulses of control message **452**. Controller **412** can perform a low pass filter operation on status signal **442** to extract symbols of feedback indicator **450**, and determine that feedback indicator **450** in the V.sub.H state. Responsive to feedback indicator **450** being in the V.sub.H state, switching cycle determination circuit **1506** sets switching cycle signal **1507** to the asserted state. In response, control signal generator **1508** generates control signals VG.sub.1 through VG.sub.4 to enable power stage **112** to transfer power to load **104** between T.sub.2 and T.sub.3. V.sub.out can also increase between T.sub.2 and T.sub.3.

(95) Also, between T.sub.3 and T.sub.5, secondary side soft start control circuit **1604** increases the target voltage code according to predetermined sequence (e.g., 0x2, 0x3, 0x4, 0x6, 0x8, 0xA, and 0xC), and feedback voltage processor **422** can adjust feedback indicator **450** to reflect the increase in the target voltage, which causes controller **412** to adjust the duty cycle of switching cycle signal **1507** (and on-time period of power stage **112**). Accordingly, V.sub.out increases between T.sub.3 and T.sub.5. V.sub.out also increases above the lower limit of the target voltage range (UVP) at T.sub.4, which causes decision signal **732** (represented by graph **2012**) to trip. If V.sub.out is still below of an upper limit of the target range, or secondary side soft start controller **1604** has not completed sequencing through the target voltage codes, status signal generator **426** can transmit the power bad message in status signal **442**. Controller **412** can also provide a power good signal **418** at a second state (e.g., a de-asserted state) to indicate that V.sub.out is still outside the target range.

(96) Referring again to FIG. 19, in step **1926**, if V.sub.out is within the target voltage range, control message generator **424** can proceed to step **1928** to end the secondary side soft start sequence, and then proceed to step **1930** to transmit a power good message in status signal **442**. Controller **412** can provide a power good signal **418** at output voltage status terminal **416** responsive to receiving the power good message, and if neither timer **1516** or timer **1518** has expired, in step **1932**. But if timer **1516** has expired (e.g., feedback indicator **450** remaining in a particular state over a threshold duration and not interrupted by a heartbeat message), or if timer **1518** has expired (e.g., power good message arriving too late after enable signal **419** is asserted), controller **412** can disable power stage **112**.

(97) The operations described above can be illustrated in FIG. 20. Specifically, at time T.sub.5, V.sub.out is within the target voltage range, and secondary side soft start controller **1604** has completed sequencing through the target voltage codes. Accordingly, secondary side soft start control circuit **1604** can assert the internal flag (labelled "Done") at T.sub.5 to indicate that the secondary side soft start operation has ended. Also, status signal generator **426** can transmit status signal **442** including the power good message. Responsive to the power good message, controller **412** can also set power good signal **418** to a first state (e.g., an asserted state) at time T.sub.6 to indicate that V.sub.out is within the target voltage range, which allows the host to enable power gate **430** and allow load **104** to draw power from secondary side coil **114b**.

(98) FIG. 21A and FIG. 21B are flowcharts of example method **2100** of operating a power converter, such as power converter **400** of FIG. 4. Method **2100** can be performed by, for example, various components of primary side circuit **402** including controller **412** and signal status receiver **414**.

(99) In step **2102**, controller **412** may receive status signal **442** via communication channel **440** between primary side circuit **402** and secondary side circuit **404** of power converter **400**. Status signal **442** may include feedback indicator **450** and control message **452**. Feedback indicator **450** can represent a relationship between an output voltage (V.sub.out) of the power converter secondary side circuit and target voltage **606**, and control message **452** can indicate an operation status of the power converter including whether the output voltage is within a target voltage range.

(100) Specifically, status signal **442** can include feedback indicator **450** within a first signal period

and control message **452** within a second signal period, and both can be in the form of symbols. Depending on the property of communication channel **440**, status signal **442** can be in the form of an electrical signal, a magnetic signal, or an optical signal. Also, status signal **442** can include a carrier signal modulated by the symbols of feedback indicator **450** and control message **452**, as shown in FIG. **14**.

(101) In some examples, feedback indicator **450** may be in the form of a hysteretic control signal, and a state of feedback indicator **450** can indicate whether V.sub.out exceeds target voltage **606**. In some examples, feedback indicator **450** can also include symbols representing a code, where the code indicates a difference between V.sub.out and target voltage **606**.

(102) Also, control message **452** can represent multiple types of control messages, each control message can represent a status of the output voltage and/or other operation conditions of power converter **400**. For example, control message **452** can represent a power good signal which indicates that V.sub.out is within a target voltage range for safe/proper operation of load **104**. Control message **452** can also represent a power bad signal which indicates that V.sub.out is outside the target voltage range. Also, control message **452** can represent a fault signal which indicates that V.sub.out is outside the target voltage range, or a temperature of secondary side circuit **404** (or power converter **400** as a whole) exceeds a threshold temperature, for a threshold duration of time. Control message **452** can include pulses in the symbols, where different types of control message **452** can be represented by different number of pulses, and the pulses can have the same pulse width between the symbols. For example, as illustrated in FIGS. **10-12**, the symbols of a power good message can have one pulse followed by an end symbol, the symbols of a power bad message can have two pulses, and the symbols of a fault message can have three pulses.

(103) In step **2104**, controller **412** can determine whether a first timer (e.g., timer **1516**) has expired when the status signal is received. Controller **412** may monitor for activity in communication channel **440** to determine whether secondary side circuit **404** is operational. Timer **1516** can expire after a threshold duration of time, which can represent a heartbeat interval. Controller **412** can start/reset timer **1516** responsive to detecting a state change in status signal **442**, which can be due to state change in feedback indicator **450**, or presence of control message **452**. If timer **1516** has expired, which can indicate potential issue at secondary side circuit **404**, controller **412** can proceed to step **2106**, where controller **412** can disable power stage **112** (e.g., by de-asserting control signals VG.sub.1 through VG.sub.4) and set output voltage status terminal **416** is set to a second state (e.g., a de-asserted state).

(104) If the timer **1516** has not expired, controller **412** can reset timer **1516**, and then proceed to step **2108** to determine a state of feedback indicator **450**. Specifically, signal extractor **1502** can detect symbols of feedback indicator **450** from status signal **442** by, for example, detecting presence or absence of a carrier signal in status signal **442**, or detecting the amplitude of status signal **442**, and synchronizing the detected symbols with clock signal **1512**. Controller **412** can also apply a low pass filtering operation to separate the symbols of feedback indicator **450** from the symbols of control message **452**, based on the symbols of feedback indicator **450** having a lower bandwidth than the symbols of control message **452**.

(105) In step **2110**, controller **412** can set an on-time period and an off-time period of power stage **112** of primary side circuit **402** based on a state of feedback indicator **450**. Specifically, referring again to FIG. **20**, in a case where feedback indicator **450** includes a hysteretic control signal, the state of feedback indicator **450** can be one of two states (e.g., V.sub.H and V.sub.L states of FIGS. **10-12**). Controller **412** can set the on-time/off time of power stage **112** (e.g., by providing switching cycle signal **1507** in the first state) responsive to feedback indicator **450** having a first state indicating that V.sub.out is below the target voltage, and the on-time period of power stage **112** can be equal to the duration in which feedback indicator **450** is in the first state. Also, controller **412** can also disable/turn off power stage **112** (e.g., by providing switching cycle signal **1507** in the second state) responsive to feedback indicator **450** having a second state indicating that

V.sub.out is above the target voltage, and the off-time period of power stage **112** can be equal to the duration in which feedback indicator **450** is in the second state. In some examples, feedback indicator **450** can represent a code representing the error/difference between V.sub.out and the target voltage, and controller **412** can set the durations of the on-time and off-time periods of power stage **112** based on the code.

(106) In step **2112**, controller **412** can determine a type of control message included in control message **452**. As described above, controller **412** can also apply a low pass filtering operation to separate the symbols of feedback indicator **450** from the symbols of control message **452**.

Controller **412** can determine a type of control message based on a number of pulses included in the symbols. For example, the symbols of a power good message can have one pulse, the symbols of a power bad message can have two pulses, and the symbols of a fault message can have three pulses. The symbols of each symbol can also include an end symbol following the pulses.

(107) In step **2114**, controller **412** can determine whether control message **452** includes a power good message based on whether control message **452** includes a single pulse followed by an end symbol. In some examples, if control message **452** includes a power good message, controller **412** can proceed to step **2115** and set output voltage status terminal **416** to a first state (e.g., by providing power good signal **418** having the first state) to indicate that V.sub.out is within the target voltage range for safe/proper operation of load **104**. Responsive to output voltage status terminal **416** being in the first state, a host can enable power gate **430** to allow load **104** to draw power from power converter **400**.

(108) In some examples, controller **412** can also determine whether the power good message is received after a second timer (e.g., timer **1518**) has expired, in step **2116**. Controller **412** can start/reset timer **1518** after receiving enable signal **419**, and determine whether the power good message is received before timer **1518** expires. If the power good message is not received before timer **1518** expires, controller **412** can also proceed to step **2116** and set output voltage status terminal **416** to the second state and disable power stage **112**.

(109) In step **2118** controller **412** determines whether a power bad message is received. Reception of the power bad message can indicate that secondary side circuit **404** is active and V.sub.out is outside the target voltage range. If the power bad message is received, controller **412** can proceed to step **2120** and set output voltage status terminal **416** to the second state, which can indicate that the secondary voltage V.sub.out is not in the target voltage range. Responsive to output voltage status terminal **416** being in the second state, the host can disable power gate **430** to prevent load **104** from drawing power from power converter **400**.

(110) If controller **412** determines that power bad message is not received, controller **412** may determine that control message **452** includes a fault message, based on whether control message **452** includes three pulses followed by an end symbol. If a fault message is received, controller **412** can also proceed to step **2106** and disable power stage **112** to stop the power transfer to secondary side circuit **404**.

(111) FIG. **21B** illustrates additional steps of method **2100** after controller **412** disables power stage **112** in step **2106**, to delay the primary side soft start sequence until the V.sub.out voltage has dropped below the UVLO threshold and to reduce the occurrence of a lockup event. Referring to FIG. **21B**, in step **2122**, controller **412** can determine whether enable signal **419** is received. Enable signal **419** can be received at terminal **417** and can be provided by the host to enable power converter **400**. Controller **412** can continue monitoring for enable signal **419** responsive to not receiving the signal.

(112) If controller **412** receives enable signal **419** in step **2122**, controller **412** can proceed to step **2124** and reset a timer (e.g., timer **1528**), to start a time window in which controller **412** monitors for status signal **442**. The reception of status signal **442** can indicate that secondary side circuit **404** is still operating, and a large residual V.sub.out voltage may remain. Accordingly, if controller **412** determines that status signal **442** is received in step **2126**, it can proceed to step **2128** and maintain

power stage **112** in the disabled state. Controller **412** can also proceed back to step **2124** to restart the timer and extend the time window. Also, if no status signal **442** is received, but the timer has not yet expired (in step **2130**), controller **412** can proceed back to step **2126** to continue monitoring for status signal **442**.

(113) On the other hand, if the timer expires (in step **2130**), which can indicate that no status signal **442** is received within the time window, controller **412** can start primary side soft start operation, in step **2132**. Details of the primary side soft start operation is described in FIG. **19** and FIG. **20**. After the primary side soft start operation completes, controller **412** can proceed to step **2102** to receive status **442** signal from secondary side circuit **404** via communication channel **440**.

(114) FIG. **22** is a flowchart of an example method **2200** of operating a power converter, such as power converter **400** of FIG. **4**. Method **2200** can be performed by, for example, various components of secondary side circuit **404** including feedback voltage processor **422**, control message generator **424**, and status signal generator **426**. Method **2200** can be performed during or after the example secondary side soft start operation of FIG. **19**.

(115) In step **2202**, feedback voltage processor **422** and control message generator **424** can receive a first voltage from an output of power converter **400**. First voltage can represent $V_{sub.out}$. In some examples, first voltage can be feedback voltage **434** from voltage divider network **432**, and first voltage can be a scaled version of $V_{sub.out}$.

(116) In step **2204**, feedback voltage processor **422** can generate feedback indicator **450** representing a relationship between the first voltage and target voltage **606**. In some examples, feedback indicator **450** can include a hysteresis control signal, where feedback indicator **450** can have one of two states depending on whether the first voltage exceeds the target voltage. In some examples, feedback indicator **450** can include a code representing an error/difference between the first voltage and the target voltage.

(117) In step **2206**, control message generator **424** can generate control message **452** indicating an operation status of power converter **400**, the operation status including whether the first voltage is within a target voltage range. As described above, control message generator **424** can generate a power good message if the first voltage is within a target voltage range, a power bad message if the first voltage is outside the target voltage range, and a fault message if, for a threshold duration of time, the first voltage is outside the target voltage range or a temperature of secondary side circuit **404** (or power converter **400**) exceeds a threshold temperature. Control message generator **424** can generate symbols including different number of pulses followed by an end symbol for different types of control messages.

(118) In step **2208**, status signal generator **426** can generate status signal **442** including feedback indicator **450** in a first signal period and control message **452** in a second signal period. Status signal generator **426** can receive symbols of feedback indicator **450** and control message **452**, synchronize them with clock signal **808**, and insert the symbols into status signal **442** within the first signal period and the second signal period defined based on clock signal **808**. In some examples, status signal generator **426** can also set the voltage state of the pulses of the symbols of control message **452** to be opposite to the voltage state of the symbols of feedback indicator **450**, as shown in FIGS. **10-12**.

(119) As described above in FIG. **9**, status signal generator **426** can determine the first symbol period (to insert the symbols of feedback indicator **450**) and the second symbol period (to insert the symbols of control message **452**) based on various techniques. For example, status signal generator **426** can determine the first and second symbol periods following a predetermined schedule. In some examples, status signal generator **426** can also interrupt the transmission of feedback indicator **450** and insert the symbols of control message **452** if control message **452** has a higher priority. For example, if control message **452** includes the first power good message after the completion of the secondary side soft start operation, status signal generator **426** may interrupt the transmission of feedback indicator **420**, and transmit control message **452** including the power

good message, to avoid controller **412** disabling power stage **112** due to expiration of timer **1518**. In some examples, if status signal generator **426** determines that feedback indicator **420** remains in a particular state over a threshold duration of time, status signal generator **426** may interrupt the transmission of feedback indicator **420** and transmit control message **452** including the power bad message as a heartbeat signal, to avoid controller **412** disabling power stage **112** due to perceived inactivity of secondary side circuit **404**.

(120) In step **2210**, status signal generator **426** can transmit status signal **442** via communication channel **440** between primary side circuit **402** and secondary side circuit **404**. Depending on the property of communication channel **440**, status signal generator **426** can transmit status signal **442** as an electrical signal, a magnetic signal, or an optical signal. Status signal generator **426** can also transmit status signal **442** by modulating a carrier signal based on the symbols of feedback indicator **450** and control message **452**.

(121) FIG. **23** is a schematic diagram of an example power converter **2300** with different semiconductor dies. In some examples, power converter **2300** includes a chip package **2350** that includes a primary side circuit **402** on a first semiconductor die **2302**, a transformer **114** on a second semiconductor die **2314**, a secondary side circuit **404** on a third semiconductor die **2304**, and a communication channel device **440** on a fourth semiconductor die **2340**.

(122) Any of the methods described herein may be totally or partially performed with a computing system including one or more processors, which can be configured to perform the steps. Thus, embodiments can be directed to computing systems configured to perform the steps of any of the methods described herein, potentially with different components performing respective steps or a respective group of steps. Although presented as numbered steps, steps of methods herein can be performed at a same time or in a different order. Additionally, portions of these steps may be used with portions of other steps from other methods. Also, all or portions of a step may be optional. Additionally, any of the steps of any of the methods can be performed with modules, units, circuits, or other means for performing these steps.

(123) In this description, the term “couple” may cover connections, communications or signal paths that enable a functional relationship consistent with this description. For example, if device A provides a signal to control device B to perform an action, then: (a) in a first example, device A is directly coupled to device B; or (b) in a second example, device A is indirectly coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal provided by device A.

(124) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(125) A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described herein as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, such as by an end-user and/or a third party.

(126) Certain components may be described herein as being of a particular process technology, but these components may be exchanged for components of other process technologies. Circuits

described herein are reconfigurable to include the replaced components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the shown resistor. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series or in parallel between the same two nodes as the single resistor or capacitor.

(127) Uses of the phrase “ground voltage potential” in this description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter.

(128) Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. An apparatus comprising: a power stage having a power stage control input; and a controller having an enable input, a timer, a control input, a control output, and an output voltage status terminal, the control output coupled to the power stage control input, and the controller configured to: receive a status signal at the control input, in which the status signal includes a set of symbols representing a feedback indicator and a control message, the feedback indicator represents a relationship between an output voltage and a target voltage, and the control message indicates whether the output voltage is within a target voltage range; responsive to the feedback indicator, provide a control signal at the control output; responsive to the control message, set a state of the output voltage status terminal indicating whether the output voltage is within the target voltage range; start the timer responsive to a state change of the enable input indicating the power stage is enabled; and responsive to not receiving the control message indicating the output voltage is within the target voltage range before the timer expires, set the control signal to disable the power stage.
2. The apparatus of claim 1, further comprising a transformer having a primary side and a secondary side, in which the power stage has a power stage output coupled to the primary side, and the secondary side is configured to provide the output voltage.
3. The apparatus of claim 1, wherein the controller is configured to set a state of the control signal responsive to a state of the feedback indicator.
4. The apparatus of claim 1, wherein the controller is configured to, before receiving the status signal, set a duty cycle of the control signal following a particular sequence of duty cycles.
5. The apparatus of claim 1, wherein the controller is configured to set the control signal to a first state responsive to the feedback indicator being in a particular state for a threshold duration of time, and the power stage is disabled responsive to the control signal having the first state.
6. The apparatus of claim 1, wherein the control message includes one of: a first message indicating that the output voltage is within the target voltage range; a second message indicating that the output voltage is out of the target voltage range; or a third message indicating at least one of: that the output voltage is outside the target voltage range for a threshold duration of time, or that a temperature exceeds a threshold temperature for the threshold duration of time.
7. The apparatus of claim 6, wherein the controller is configured to: set the output voltage status terminal to a first state responsive to the control message including the first message; and set the output voltage status terminal to a second state responsive to the control message not including the first message.
8. The apparatus of claim 6, wherein the controller is configured to set the control signal to a first state responsive to the control message including the third message, and the power stage is disabled

responsive to the control signal having the first state.

9. The apparatus of claim 8, wherein the controller is configured to: set the control signal to the first state responsive to not receiving the control message including the first message before the timer expires.

10. The apparatus of claim 8, wherein the controller has an enable input and includes a timer, and the controller is configured to: set the control signal to the first state; start the timer responsive to a state change of the enable input; reset the timer responsive to receiving the status signal; and after the timer expires, set the control signal to a second state; and wherein the power stage is enabled responsive to the control signal having the second state.

11. The apparatus of claim 6, wherein: the set of symbols includes a first set of symbols and a second set of symbols; the control message includes first, second, and third numbers of pulses representing respective first, second, and third messages, and a pulse of the first, second, and third numbers of pulses has a different state from the feedback indicator preceding the control message in the first set of symbols; and the second set of symbols includes the first, second, and third numbers of pulses.

12. The apparatus of claim 11, wherein the controller is configured to: determine a state of the feedback indicator by performing a low pass filtering operation on the status signal; and identify one of the first, second, or third messages based on a number of state transitions in the status signal.

13. The apparatus of claim 1, wherein the status signal includes an electrical signal, a magnetic signal, or a light signal.

14. An apparatus comprising: a feedback voltage processor having first and second inputs and a feedback output, the first input coupled to a power converter output terminal, the second input coupled to a target voltage terminal, and the feedback voltage processor configured to generate a feedback indicator at the feedback output responsive to a first voltage at the power converter output terminal and a target voltage at the target voltage terminal; a control message generator having a control input and a control message output, the control input coupled to the power converter output terminal, and the control message generator configured to provide a control message at the control message output responsive to whether the first voltage is within a target voltage range; and a status signal generator having a first signal input, a second signal input, and a status signal output, the first signal input coupled to the feedback output, the second signal input coupled to the control message output, and the status signal generator configured to: receive the feedback indicator at the first signal input; receive the control message at the second signal input; and provide a status signal at the status signal output, the status signal including a set of symbols that includes the feedback indicator and the control message, wherein: the set of symbols includes a first set of symbols and a second set of symbols; the control message includes first, second, and third numbers of pulses representing respective first, second, and third messages, and a pulse of the first, second, and third numbers of pulses has a different state from the feedback indicator preceding the control message in the first set of symbols; and the second set of symbols includes the first, second, and third numbers of pulses.

15. The apparatus of claim 14, wherein the status signal generator is configured to provide the feedback indicator as the status signal in a first signal period and the control message as the status signal in a second signal period.

16. The apparatus of claim 14, further comprising a transformer having a primary side and a secondary side, in which the power converter output terminal is coupled to the secondary side, and the status signal is targeted at a controller coupled to the primary side.

17. The apparatus of claim 14, wherein the feedback voltage processor is configured to: set the feedback indicator to a first state responsive to the first voltage exceeding the target voltage; and set the feedback indicator to a second state responsive to the first voltage not exceeding the target voltage.

18. The apparatus of claim 14, wherein the control message includes one of: a first message

indicating that the first voltage is within the target voltage range; a second message indicating that the first voltage is outside the target voltage range; or a third message indicating at least one of: that the first voltage is outside the target voltage range for a threshold duration of time, or that a temperature exceeds a threshold temperature for the threshold duration of time.

19. The apparatus of claim 18, wherein the status signal generator is configured to: provide the feedback indicator as the status signal; receive the control message; and responsive to the feedback indicator being in a particular state for a threshold duration of time: stop providing the feedback indicator as the status signal; and provide the control message as the status signal.

20. The apparatus of claim 18, further comprising a soft start control circuit having a status signal control output, the soft start control circuit configured to provide a control signal at the status signal control output indicating whether a soft start operation completes, wherein the status signal generator has a status signal control input coupled to the status signal control output and is configured to: provide the feedback indicator as the status signal; receive the control message including the first message; receive the control signal at the status signal control input; and responsive to receiving the first message, and the control signal indicating that the soft start operation completes: stop providing the feedback indicator as the status signal; and provide the control message including the first message as the status signal.

21. The apparatus of claim 20, wherein the control signal is a first control signal, and the apparatus further comprises a voltage source having a target voltage control input and a target voltage output, the target voltage output coupled to the target voltage terminal, and the voltage source configured to set the target voltage responsive to a second control signal at the target voltage control input, and provide the target voltage at the target voltage output; and wherein the soft start control circuit has a target voltage control output coupled to the target voltage control input and configured to: provide a sequence of the second control signal indicating different target voltages; and after providing the sequence, provide the first control signal at the status signal control output, in which the first control signal indicates that the soft start operation has completed.

22. The apparatus of claim 21, wherein the soft start control circuit is configured to, responsive to the first voltage being below an undervoltage lockout (UVLO) threshold, provide a third control signal at the status signal control output indicating that the status signal generator is to be disabled, and the status signal generator is disabled responsive to receiving the third control signal.

23. A method comprising: receiving an enable input indicating a power converter is enabled to output power; responsive to the enable input, starting a timer; receiving a status signal via a communication channel between primary side and secondary side circuits of the power converter, in which the status signal includes a set of symbols based on a feedback indicator and a control message, in which the feedback indicator represents a relationship between a target voltage and an output voltage of the secondary side circuit, and the control message indicates whether the output voltage is within a target voltage range; determining a state of the feedback indicator; responsive to the state of the feedback indicator, set an on-time period and an off-time period of a power stage of the primary side circuit; responsive to the control message, set a state of an output voltage status terminal indicating whether the output voltage is within the target voltage range; and responsive to not receiving the control message indicating the output voltage is within the target voltage range before the timer expires, set a control signal to disable the power converter.

24. The method of claim 23, wherein the control message includes one of: a first message indicating that the output voltage is within the target voltage range; a second message indicating that the output voltage is outside the target voltage range; or a third message indicating at least one of: that the output voltage is outside the target voltage range for a threshold duration of time, or that a temperature of the secondary side circuit exceeds a threshold temperature for the threshold duration of time.

25. The method of claim 24, further comprising: responsive to receiving the control message including the first message, setting the output voltage status terminal to a first state; responsive to

receiving the control message including the second message, setting the output voltage status terminal to a second state; and responsive to receiving the control message including the third message, disabling the power stage.

26. The method of claim 23, further comprising, before receiving the status signal, setting the on-time period and the off-time period of the power stage following a particular sequence of duty cycles.

27. A method comprising: receiving a first voltage from an output of a power converter; generating a feedback indicator representing a relationship between the first voltage and a target voltage; generating a control message indicating whether the first voltage is within a target voltage range; generating a status signal that includes a set of symbols based on the feedback indicator and the control message; and transmitting the status signal via a communication channel between primary and secondary side circuits of the power converter; wherein: the set of symbols includes a first set of symbols and a second set of symbols; the control message includes first, second, and third numbers of pulses representing respective first, second, and third messages, and a pulse of the first, second, and third numbers of pulses has a different state from the feedback indicator preceding the control message in the first set of symbols; and the second set of symbols includes the first, second, and third numbers of pulses.

28. The method of claim 27, wherein the control message includes one of: a first message indicating that the first voltage is within the target voltage range; a second message indicating that the first voltage is outside the target voltage range; or a third message indicating at least one of: that the first voltage is outside the target voltage range for a threshold duration of time, or that a temperature of the secondary side circuit exceeds a threshold temperature for the threshold duration of time.

29. The method of claim 27, further comprising: determining whether the first voltage exceeds a UVLO threshold; and responsive to the first voltage exceeding the UVLO threshold, increasing the target voltage based on a particular sequence of target voltages.

30. An apparatus, comprising: a primary side circuit including: a power stage having a power stage voltage input, a power stage control input, and a power stage output; and a controller having an enable input, a timer, a power stage control output, an output voltage status terminal, an enable input terminal, and a status signal input, the power stage control output coupled to the power stage control input, wherein the controller is configured to: start the timer responsive to a state change of the enable input indicating the power stage is enabled; and responsive to not receiving a control message indicating an output voltage at an output voltage terminal is within a target voltage range before the timer expires, set a control signal to disable the power stage; a transformer having a primary side coil and a secondary side coil, the primary side coil coupled to the power stage output; a secondary side circuit having a secondary side input and an the output voltage terminal, the secondary side input coupled to the secondary side coil, the secondary side circuit including: a feedback voltage processor having a feedback input and a feedback output, the feedback input coupled to the output voltage terminal; a control message generator having a control input and a control message output, the control input coupled to the output voltage terminal; and a status signal generator having first and second signal inputs and a status signal output, the first signal input coupled to the feedback output, and the second signal input coupled to the control message output; and a communication channel device having a channel input and a channel output, the channel input coupled to the status signal output, and the channel output coupled to the status signal input.

31. The apparatus of claim 30, wherein: the controller is configured to: receive a status signal at the status signal input, the status signal including a feedback indicator and a control message; generate a control signal at the power stage control output responsive to a state of the feedback indicator; and set a state of the output voltage status terminal responsive to the control message; the feedback voltage processor is configured to generate, at the feedback output, the feedback indicator representing a relationship between a first voltage at the output voltage terminal and a target

voltage; the control message generator is configured to generate the control message indicating whether the first voltage is within a target voltage range; and the status signal generator is configured to generate the status signal responsive to the feedback indicator at the first signal input and the control message at the second signal input; and the communication channel device is configured to transmit the status signal from the channel input to the channel output.

32. The apparatus of claim 30, wherein: the primary side circuit is on a first semiconductor die; the transformer is on a second semiconductor die; the secondary side circuit is on a third semiconductor die; the communication channel device is on a fourth semiconductor die; and first through fourth semiconductor dies are enclosed in a chip package.
